

SAI KIRAN POTRU

Software Engineer | Full-Stack Java Developer
Mobile: 980-483-4163 | Email: spkiran0225@gmail.com
Current Location: Charlotte, NC
LinkedIn: [linkedin.com/in/sai-kiran-potru-6a4b87263/](https://www.linkedin.com/in/sai-kiran-potru-6a4b87263/)



PROFESSIONAL SUMMARY:

- Software Engineer with 4+ years of experience building scalable backend systems, cloud-native applications, and data-driven solutions for healthcare and enterprise environments.
- Skilled in building secure and scalable RESTful APIs using Spring Boot and Python, and experienced with containerization using Docker and orchestration with Kubernetes.
- Strong background in cloud platforms, leveraging AWS services (Lambda, Glue, S3, Redshift, EC2, RDS) and Microsoft Azure services (Functions, Blob Storage, Virtual Machines, SQL Database, Event Hubs) along with CI/CD automation to improve scalability and reduce release cycles.
- Experienced in real-time data streaming and processing with Apache Kafka and Apache Spark, along with data warehousing and analytics using Snowflake and PostgreSQL.
- Proficient in Infrastructure as Code (Terraform) for reliable resource provisioning and monitoring solutions using ELK Stack, Prometheus, and Grafana.
- Adept at writing unit and integration tests with Postman, JUnit, Mockito, and PyTest to ensure high-quality, reliable deliverables.
- Collaborative team player experienced in Agile/Scrum, working with cross-functional teams to provide high-quality features on time.
- Robust problem-solving and communication skills, with a track record of translating business needs into technical solutions that enhance efficiency and compliance.

TECHNICAL SKILLS:

Category	Skills & Tools
Programming Languages	Java, Python, C++, JavaScript, TypeScript, SQL, HTML, CSS, Bash
Frameworks & Libraries	Spring Boot, React, Angular, ASP.NET Core, Hibernate, Entity Framework, Bootstrap, jQuery
APIs & Protocols	RESTful APIs, Swagger/OpenAPI, OAuth2, JWT
Security & Compliance	AES Encryption, TLS/SSL, RBAC, HIPAA Compliance
Cloud Platforms & Services	AWS (Lambda, Glue, S3, EC2, RDS, Redshift, CloudWatch, IAM), Microsoft Azure (Functions, ADF, Blob Storage, Virtual Machines, SQL Database, Event Hubs, Monitor, AD)
Data Engineering & Processing	Apache Kafka, Apache Spark, ETL Workflows, Snowflake, Data Lakes, PostgreSQL
Containerization & Orchestration	Docker, Kubernetes
IaC & Data Visualization	Terraform, Power BI
CI/CD & Version Control	Jenkins, Git, GitFlow, GitHub Actions, Azure DevOps
Monitoring & Observability	ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana
Testing Tools	Postman, JUnit, Mockito, PyTest
Methodologies & Practices	Agile/Scrum, Code Reviews, API & System Documentation
Machine Learning & Data Handling	ML Model Deployment, Data Preprocessing, Structured/Semi-Structured/Unstructured Data Integration

PROFESSIONAL EXPERIENCE

McKesson — Software Engineer (Data & Backend Focus) **Charlotte, NC — Aug 2023 – Present**

- Developed RESTful APIs with Spring Boot and Python for HIPAA-compliant healthcare systems, and reduced data errors by

25%.

- Applied AES, TLS/SSL, JWT, and RBAC to protect patient data and maintain compliance.
- Migrated ETL workflows to AWS Lambda and Glue, cutting runtime by 40% and lowering costs by 20%.
- Built streaming pipelines with Apache Kafka, enabling high-throughput ingestion with sub-second latency.
- Designed and implemented data pipelines to integrate structured and semi-structured healthcare data.
- Optimized Snowflake schemas, indexes, and queries, improving reporting speed by 60% and reducing compute usage by 15%.
- Used AWS services (S3, EC2, RDS, CloudWatch, IAM) for secure and scalable deployments.
- Applied Apache Spark for large-scale data processing and analytics.
- Built data lake and warehousing solutions with AWS Redshift to support reporting and analytics.
- Containerized applications with Docker, deployed to Kubernetes, enable auto-scaling and rolling updates.
- Automated build, test, and release pipelines with Jenkins CI/CD, reducing release cycles by 30%.
- Provisioned cloud resources using Terraform (IaC), reducing setup time by 40%.
- Integrated monitoring and auditing with the ELK Stack (Elasticsearch, Logstash, Kibana).
- Tested services with Postman, JUnit, Mockito, and PyTest, improving quality and reducing defects.
- Managed source control with Git and GitFlow branching strategies to support team collaboration.
- Worked in Agile/Scrum teams, coordinating with product owners, QA, and data scientists to deliver features on schedule.
- Partnered with data scientists to deploy machine learning models into production pipelines, ensuring scalability and compliance.
- Built data preprocessing workflows for ML training datasets, handling large volumes of structured and unstructured healthcare data.

Adani — Software Developer

India — May 2020 – Aug 2022

- Developed and upgraded internal web applications using Java, Spring Boot, and React, migrating legacy systems to a scalable microservices architecture.
- Designed and implemented RESTful APIs to support secure and efficient data exchange across business systems.
- Automated reporting with Python, reducing report generation time from hours to minutes.
- Optimized PostgreSQL queries and added indexes, improving database performance by 35%.
- Built and maintained Jenkins CI/CD pipelines, ensuring consistent and reliable deployments.
- Applied GitFlow branching strategies and peer code reviews to maintain high code quality and streamline team collaboration.
- Containerized applications with Docker to ensure consistent deployments across development, testing, and production environments.
- Authored and maintained API documentation with Swagger/OpenAPI, simplifying integration for internal and external teams.
- Enhanced security by executing OAuth2 authentication and role-based access controls (RBAC).
- Deployed services on Microsoft Azure (Virtual Machines, Blob Storage, Azure Functions), enabling scalable, cost-efficient, and resilient solutions.
- Improved system reliability by monitoring applications with Prometheus, Grafana, and log-based analysis, and reduced downtime and accelerated issue resolution.
- Created and maintained unit and integration tests with JUnit and Mockito to improve test coverage.
- Worked in Agile/Scrum sprints with product owners, QA, and developers to provide features on schedule.
- Collaborated with stakeholders to gather requirements and align development with business goals.
- Prepared clear documentation of system designs, deployment processes, and workflows to support future enhancements.

EDUCATION:

M.S. in Computer Science

Northern Arizona University, 2022–2023

B.Tech, Information Technology

KL University, 2018–2022

CERTIFICATIONS:

- AWS Solution Architect Professional
- Microsoft Azure Developer Associate